Derivable Judgement

A Statistical Decision-Making Model — Theory Report (Part A)

Context: A data analyst at a public health research organization uses inferential statistics on a **600-record health dataset** (age, weight, gender, region, smoking status, exercise, BMI, blood pressure, cholesterol, glucose, diabetes, hypertension) to derive judgements about factors affecting **diabetes** and **hypertension**. All tests use a significance level **α = 0.05**.

1. What is Inferential Statistics?

Inferential statistics draws conclusions about a population from a sample. It uses estimation (point estimates and confidence intervals) and hypothesis testing, and always quantifies the uncertainty of the conclusion.

> *In this project: the 600 records are a sample; we infer which factors (smoking, age, gender) affect disease in the wider population.*

2. Hypothesis Testing and its Components

Hypothesis testing is a formal procedure to decide between two competing claims using sample data. Components: **Null hypothesis (H_0)** — no effect/difference; **Alternative hypothesis (H_1)** — an effect/difference exists; **Significance level (α)** — tolerated probability of a false rejection (commonly 0.05); **Test statistic** — value computed from the sample; **Critical value / p-value** — basis for the decision; **Decision** — reject or fail to reject H_0.

> *In this project: H_0: smoking has no effect on diabetes prevalence; H_1: smoking affects diabetes prevalence.*

3. Confidence Interval and Critical Value

A confidence interval is a range of values likely to contain the true population parameter at a stated confidence level (e.g. 95%). The critical value is the cut-off point on the test distribution (z, t, χ^2 or F) that marks the rejection region; if the test statistic falls beyond it, H_0 is rejected.

> **Formula: CI = x-bar ± (critical value) × (s / √n)**
>
> *In this project: 95% CI for mean systolic BP is 108.73 to 110.44 mmHg (mean 109.59), a narrow interval because n = 600.*

4. p-value

The p-value is the probability of obtaining a result at least as extreme as the one observed, assuming H_0 is true. A small p-value ($p < \alpha$) is evidence against H_0, so H_0 is rejected; a large p-value means we fail to reject it.

> *In this project: smoking vs diabetes gives p = 0.566 (fail to reject), while diabetes across age groups gives p < 0.001 (reject).*

5. Type I vs Type II Errors

Type I error (α) — rejecting H_0 when it is actually true (false positive). **Type II error (β)** — failing to reject H_0 when it is actually false (false negative). Reducing one generally increases the other; power of a test = $1 - \beta$.

> *In this project: a Type I error would mean wrongly concluding that smoking raises diabetes risk; a Type II error would mean missing a real effect.*

6a. z-test

Used to compare a sample mean/proportion with a population value, or two means, when the sample is large ($n \geq 30$) or the population standard deviation is known. The statistic follows the

standard normal distribution.

Formula: $z = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$

In this project: H3 compares mean BMI of males (n = 321) and females (n = 251); both groups are large, so a z-test is used.

6b. t-test

Used to compare means when the sample is small or the population standard deviation is unknown. Types: one-sample, independent two-sample (Student/Welch) and paired. The statistic follows the t-distribution with the appropriate degrees of freedom.

Formula: $t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$

In this project: H2 uses the Welch t-test (unequal variances) to compare mean BP of smokers and non-smokers.

6c. Chi-square test

Tests whether two categorical variables are associated (test of independence) or whether observed frequencies fit an expected distribution (goodness-of-fit), by comparing observed and expected counts.

Formula: $\chi^2 = \Sigma\ (O - E)^2 / E$

In this project: H1 tests the association between smoking status (3 categories) and diabetes (yes/no), with df = 2.

6d. ANOVA

Analysis of Variance compares the means of three or more groups simultaneously, avoiding the inflated Type I error of many t-tests. It uses the F-statistic: the ratio of between-group variance to within-group variance.

Formula: F = MS(between) / MS(within)

In this project: H4 compares the diabetes rate across five age groups (18-25, 26-35, 36-45, 46-60, 60+), with df = (4, 595).

7. Covariance

Covariance measures how two variables change together. Positive covariance means they tend to rise together; negative means one rises as the other falls. Its magnitude depends on the units, so it is hard to compare across variable pairs.

Formula: $\mathrm{Cov}(X, Y) = \Sigma\ (x_i - \bar{x})(y_i - \bar{y}) / (n - 1)$

In this project: the covariance matrix of age, weight, BMI, BP, cholesterol and glucose shows the direction of each pairwise relationship.

8. Correlation

Correlation (Pearson r) is the standardised covariance. It ranges from −1 (perfect negative) to +1 (perfect positive linear relationship); 0 means no linear relationship. Correlation does not imply causation.

Formula: $r = \mathrm{Cov}(X, Y) / (s_x \cdot s_y)$

In this project: H5 tests the linear relationship between age and BMI; the heatmap shows all pairwise correlations.

Summary of Test Selection

Question	Variables	Test
Do two group means differ (large n)?	Numeric vs 2 groups	z-test
Do two group means differ (small n)?	Numeric vs 2 groups	t-test
Are two categories associated?	Categorical vs categorical	Chi-square
Do 3+ group means differ?	Numeric vs 3+ groups	ANOVA
Are two numeric variables related?	Numeric vs numeric	Covariance / Correlation

Test Selection Used in This Project

Hypothesis	Variables	Test used
H1: Smoking vs diabetes	Smoking status vs diabetes (categorical)	Chi-square
H2: BP of smokers vs non-smokers	Blood pressure vs smoking group (2 groups)	Welch t-test
H3: BMI of males vs females	BMI vs gender (2 groups, large n)	Two-sample z-test
H4: Diabetes rate across age groups	Diabetes vs age group (5 groups)	One-way ANOVA
H5: Age and BMI relationship	Age vs BMI (numeric vs numeric)	Pearson correlation